AIeveR



Spark 2

User Manual

Robotic Arm · Safety · Installation · Web GUI · Teach · Maintenance

Read all safety instructions before operating the robot.

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1 Safety Instructions

⚠ Important

This section is a condensed set of critical safety rules. Always follow site-specific procedures and applicableregulations. Contact technical support for abnormal conditions; do not attempt unauthorized repairs.

Operating Environment:
Temperature -20°C to 50°C, relative humidity 25% to 85% (non-condensing)

Power Supply:
Standard DC 48V

Payload:
End-effector payload ≤ 1.9 kg

Workspace:
Ensure there is sufficient clear space around the robotic arm before use;
personnel must remain outside the operating range during operation

Attire:
Do not wear loose clothing during operation. Tie back long hair,
and wear safety goggles when entering the work area

Temperature:
The robotic arm may generate heat during operation or immediately after shutdown;
it is recommended to power off and allow the device to cool for 1 hour before touching

Magnetic Fields:
Avoid prolonged exposure to strong magnetic fields

Modifications:
Unauthorized modifications to the robotic arm are prohibited

Target Users:
Operation by individuals under 18 years of age or those without full civil capacity is prohibited

Abnormalities:
In the event of an abnormality, stop operation immediately and contact technical support.
Do not attempt to handle the issue without authorization

2 Product Overview

Spark2 is designed as a research and education platform for robotics and AI. It features a compact structure and integrated force control, enabling precise motion execution and compliant physical interaction. These capabilities make it suitable for fine manipulation, human–robot collaboration, andalgorithm development in space-constrained or unstructured environments. With open interfaces andeasy deployment, the system supports rapid prototyping, experimentation, and integration withmodernAI models.

2.1 Product List

Charger



AC/DC Switching Adapter

Circular Connector



Ethernet Cable



Spark 2



Gripper

2.2 Technical Specifications

Degrees of Freedom	6
Payload	1.9kg
Weight	~6kg
Working Reach	723mm
Repeatability Accuracy	/
Standard Supply Voltage	DC 48V
Material	Aluminum frame + Plastic shell
Communication Method	CAN Communication
Control Method	Teach and Play/GUI/SDK
External Interfaces	Power ×1, Network ×1
Base Mounting	100 × 100 mm, M6 × 4
IP Rating	IP54
Operating Environment	-20~50°C, 25%~85% RH (non-condensing)
Noise Level	<60 dB

2.3 Joint Movement Range

	Motion Range	Velocity Range
J1	±180°	300°/s
J2	±90°	300°/s
J3	±180°	300°/s
J4	-20°-180°	300°/s
J5	±180°	300°/s
J6	±360°	300°/s

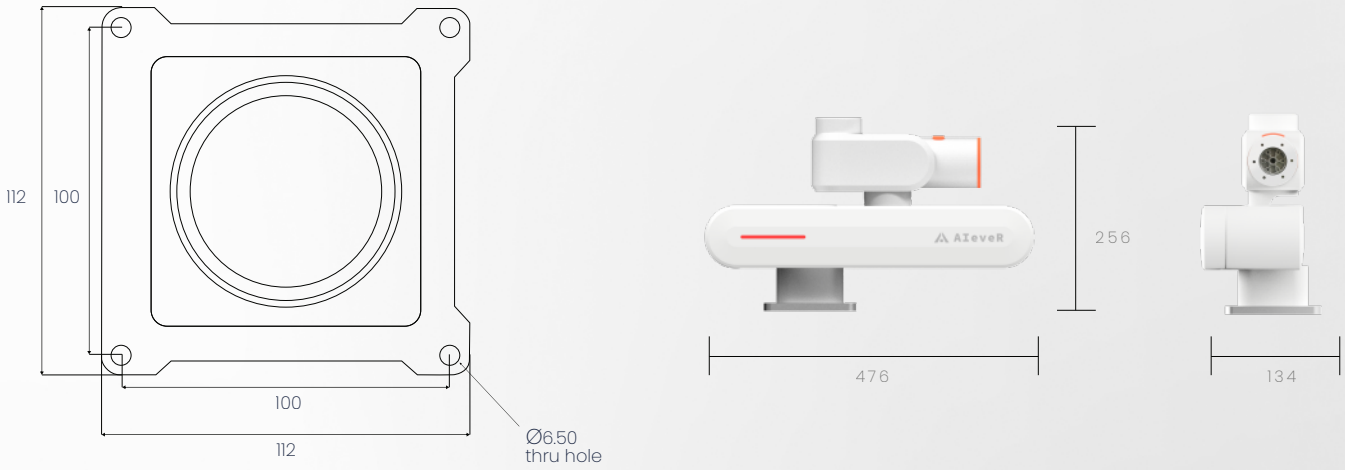
2.4 Modified DH

	α_{i-1}	a_{i-1}	θ_i	d_i
J1	0	0	θ_1	0.0982
J2	-90	0.096	θ_2 -90	0
J3	0	0.3465	θ_3	0
J4	0	0.15	θ_4	0
J5	90	0	θ_5 +90	0.09572
J6	90	0	θ_6	0.1189

3 Installation

3.1 Mechanical Installation

The robotic arm base features four M6 threaded through-holes, with a hole spacing of 100 mm.



*mm

Figure 3-1 Base mounting dimensions

3.2 Electrical Installation

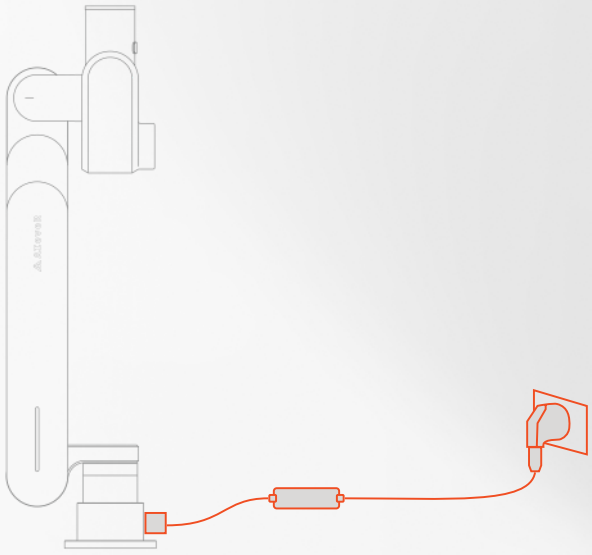


Figure 3-2 Electrical connector reference (ports A / B)

Operation

1 Insert the aviation plug at Position A according to the connector orientation
2 Connect the B-end of the power adapter to the mains power supply

> **Caution**
> Verify voltage and polarity before powering on. Use only the approved power adapter. Do not hot-plug connectors while the system is powered.

3.3 Using the Aviation Connector

The aviation plug and the robot arm base each have a red mark.
Line the two red marks up, then push the plug straight in (Figure 1 and Figure 2).
To remove the plug, pinch the front half and pull it backward, then pull the plug out (Figure 3).

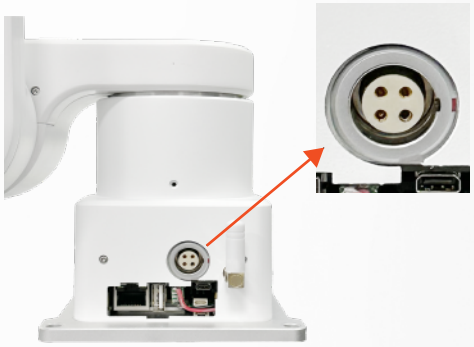


Figure 1

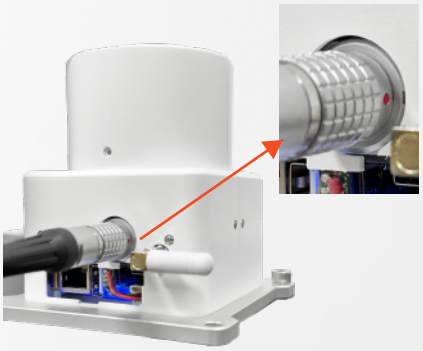


Figure 2

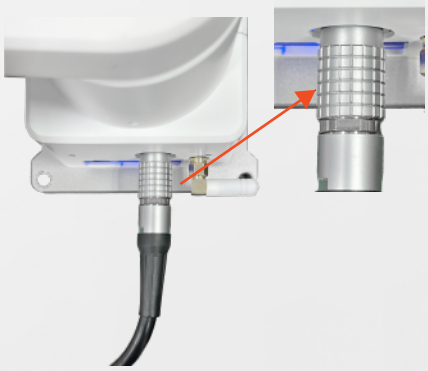


Figure 3

> **WARNING**
> Do not connect or disconnect the aviation connector while the system is powered on.

3.4 Gripper Installation

Installation Procedure :



Figure 1

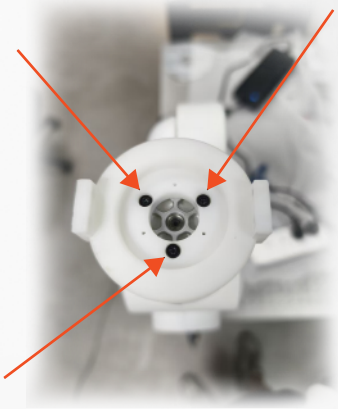


Figure 2

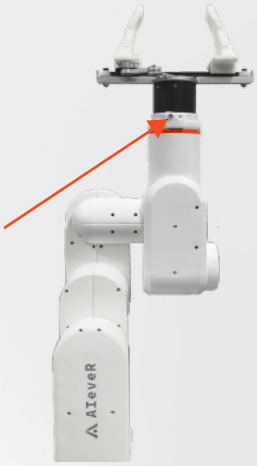


Figure 3



Figure 4

1. Align the locating slot on the flange with the locating slot on the J6 motor, as shown in Figure 1. Secure the flange using three M3 screws. The tightened assembly should look as shown in Figure 2. (The robot arm must be in its zero position.)

2. Align the locating slot on the bottom of the gripper with the locating slot on the flange, and secure the gripper using four M3 screws, as shown in Figure 3. Make sure that the signal cable on the gripper is aligned with the corresponding connector on the robot arm, as shown in Figure 4. (The robot arm must be in its zero position.)

4 Coordinate Systems and Home Position

The coordinate origin of Spark2 is located at the center of the base. The coordinate axes and zero positions of each joint are shown in the figure.

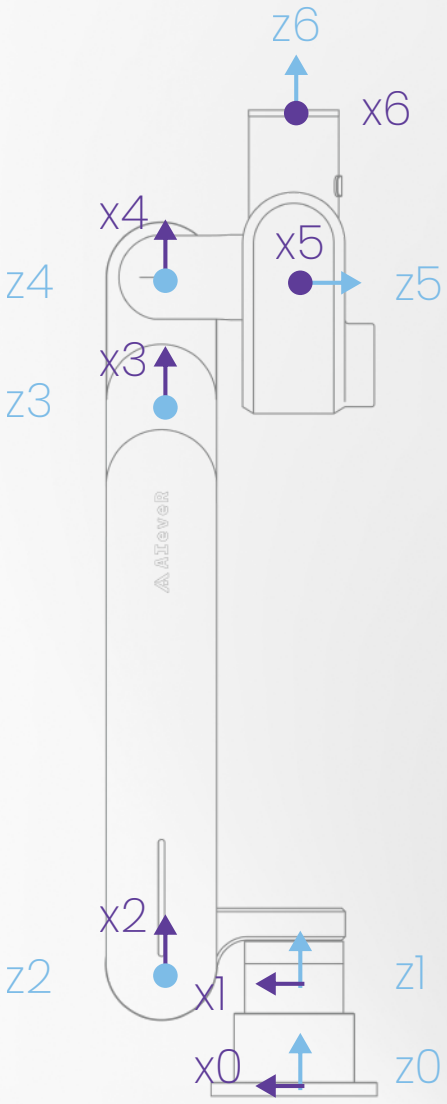


Figure 4-1 Coordinate systems and joint zero positions

5 Drag Teaching (Physical Teach Button)

The teach button is read through the CAN button module. It is used to enter teach mode for manual guiding, and to trigger recovery when the system is in a fault state.

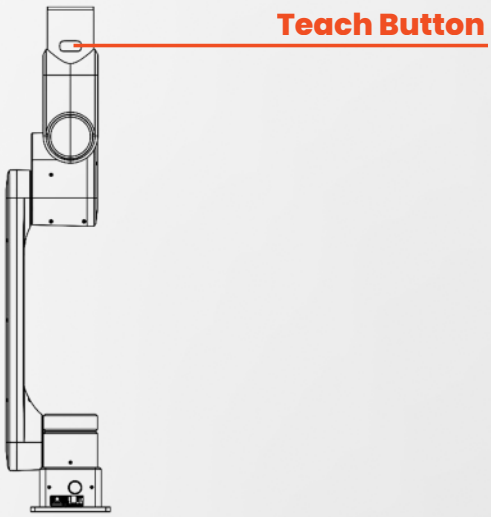


Figure 7-1 Teach button

5.1 Normal Operating Condition

The normal operating condition means that the robot arm is not in a fault state.

Short Press:

- The robot arm enters teach mode and can be guided manually, subject to the configured safety checks.
- The teach button acts as a local, high-priority control client (client ID: 10001), with a higher priority than normal clients.

Release:

- Releasing the teach button relinquishes control, and the robot arm enters a controlled stop or idle state. Other clients may regain control according to their priority, including:
    - SDK
    - Panel Web Client

5.2 Fault / Error Condition

A fault condition means that the system is already in a fault state (e.g., a hard-limit violation, self-collision, invalid configuration, or repeated hardware failures). In this state:

- Normal motion commands are rejected. A short press of the teach button will not activate teach mode.

Long Press:

- A recovery request is sent, and the system enters Recovery mode.
- If the safety checks pass, the fault is cleared. Otherwise, the system remains in the fault/error state.

Release:

- Once the fault has been cleared, the robot arm returns to the normal operating condition.

5.3 Notes

1. Recovery is not a bypass: if the hazard remains, the system stays faulted or returns to error. Recoveronly after it is safe.
2. Do not use the teach button as an emergency stop; use a dedicated physical E-stop.
3. If button hardware is configured but communication fails repeatedly, the systemraises kFaultButtonNotFound.

6 Safety Page — Fault Conditions

Once faulted, normal commands are rejected until a valid recovery request is processed and safety checks pass.

6.1 Relevant System States

State	Meaning
Error	A safety fault has been detected
Recovery	Recovery behavior is active
Offline	Required hardware communication failed; hardware considered disconnected

Related diagnostic while faulted: **kFaultRecoveryRequired**

6.2 Fault Conditions (Require Recovery)

Hard Limits

Condition	Diagnostic	Trigger
Arm hard position limit	kFaultPosHardLimitReached	Measured joint position outside hard positionlimit (lower or upper)
Arm hard velocity limit	kFaultVelHardLimitReached	Measured joint velocity outside hard velocity limit
Arm hard torque limit	kFaultTorHardLimitReached	Measured joint torque outside hard torque limit

Operator action: Stop motion immediately. Move away via approved recovery only after the cause is understood and the arm is stationary / safe.

Tracking Failures

Condition	Diagnostic	Trigger
Position tracking failure	kFaultPosTrackingFailed	In position control, | commanded - measured | exceeds follow limit
Velocity tracking failure	kFaultVelTrackingFailed	In velocity control, | commanded - measured | exceeds follow limit
Torque tracking failure	kFaultTorTrackingFailed	In torque control, | commanded - measured | exceeds follow limit

Typical causes: blocked / overloaded joint, aggressive command, motor not enabled or saturated, mechanical obstruction, incorrect feedback / zero offset, external contact.

Self-Collision

Condition	Diagnostic	Trigger
Position tracking failure	kFaultCollisionDetected	Self-collision detection enabled and checker reports link–link collision

System state is set to Error.

Operator action: Stop all motion; identify colliding links from diagnostics/logs; use recovery only tomoveaway from the collision pose; verify self_collision_detection settings before resuming.

Hardware Not Found / Disconnected / Enable / Mode / I/O

Condition	Diagnostic	Trigger
Arm not found / init failed	kFaultArmNotFound, kFaultArmInitFailed	Cannot open arm, or communication fails repeatedly
Gripper not found / init failed	kFaultGripperNotFound, kFaultGripperInitFailed	Cannot open gripper, or command/read fails repeatedly
Button not found	kFaultButtonNotFound	Teach button expected but communication fails repeatedly
Hardware enable failed	kFaultHardwareEnableFailed	Cannot enable joints or update motor enable state
Hardware mode change failed	kFaultHardwareChangeModeFailed	Cannot switch to requested actuator mode
Hardware set/read failed	kFaultHardwareSetReadFailed	Setting command or reading feedback fails

Operator action: Check power, wiring, permissions, drive/brake/fault status, and configuration; reconnect or restart hardware only when safe.

Invalid Robot / Message / Control Configuration

Condition	Diagnostic	Trigger
Robot / size mismatch	kFaultRobotNameMismatch, kFaultArmSizeMismatch, kFaultArmJointSizeMismatch, kFaultGripperSizeMismatch, kFaultGripperJointSizeMismatch	Client request does not match server configuration
Invalid control configuration	kInvalidControlTypeCombination, kCartesianControlRequirePositionTarget, kControlStrategyNotAvailable, kPlaybackControlRequirePositionTarget, kWaypointSmoothingMethodNotAllowed, kWaypointControlStrategyNotAllowed, kWaypointTargetTypeNotAllowed	Requested strategy / target type / modenot supported

Operator action: Reload client config from the server; correct control mode; do not force commands from a mismatched client.

External Safeguard and Unknown Fatal

Condition	Diagnostic	Trigger
External safeguard request	Monitoring interface safeguard	Valid session requests protective stop →faulted
Unknown fatal exception	kUnknown	Unexpected exception during startup or runtime

Operator action: Treat safeguard as a safety stop and check the external monitor/client. For kUnknown: stop the arm, save logs, restart only after investigation.

7 Maintenance and Troubleshooting

7.1 Storage

- After use, return the robot arm to the zero position, power it off, and store it properly.
- Organize and bundle the cables to prevent tangling or excessive pulling.
- Storage environment: Temperature -20°C to 50°C, humidity 25% to 85% (non-condensing).

7.2 Periodic Maintenance

> **Note**
> All maintenance must be performed with the robot arm at the zero position and powered off.

Item	Procedure
Robot Arm Cleaning	Wipe off dust, dirt, and grease using a wet wipe or a wrung-out damp cloth.
Screw Fastening	Inspect and tighten all external screws if loose.
Motor Inspection	Check the motors and structural components for any damage or deformation.
Cable Inspection	Check that the power cable connection is secure and the outer sheath is free from damage.

8 Dimensional Drawings

8.1 Arm

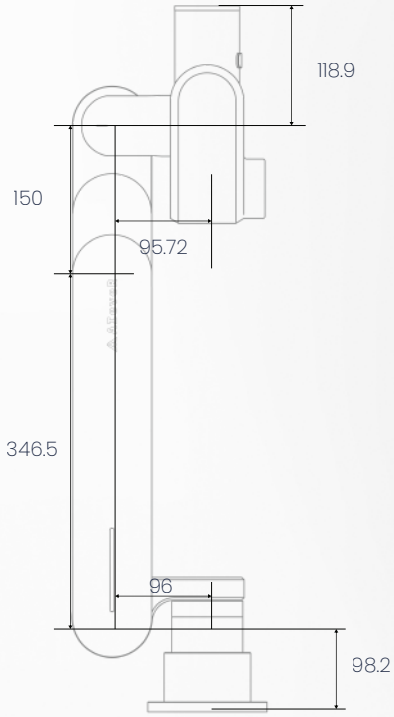


Figure 9-1 Arm dimensions

8.2 Flange

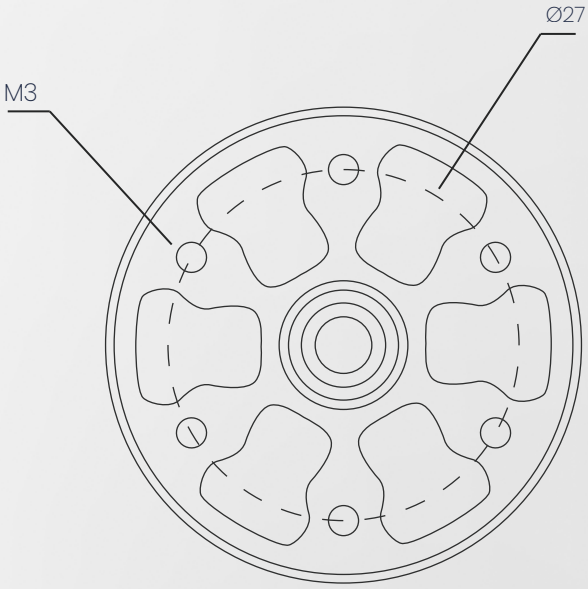


Figure 9-2 Flange

9 Disclaimer

This content is provided for operational guidance for the cuarm_rt_control software and does not replace formal safety validation, risk assessment, hardware manuals, or site-specific operating procedures.

The software includes diagnostic checks, command limiting, collision checking, and recovery behavior, but these mechanisms cannot guarantee safe operation under all conditions. Robot safety depends oncorrect mechanical installation, electrical wiring, hardware configuration, calibration, operator training, workspace design, emergency-stop integration, and ongoing maintenance.

The authors, maintainers, and affiliated organizations are not liable for injury, death, equipment damage, data loss, production loss, or any direct or indirect damages resulting from use, misuse, modification, integration, or failure of this software or the robot system.

Operators and integrators are responsible for:

- Performing a complete risk assessment before use.
- Verifying all limits, collision models, tool offsets, payloads, and recovery procedures.
- Ensuring compliance with applicable laws, regulations, standards, and institutional policies.
- Providing independent hardware-level safety devices where required.
- Training all users before allowing robot operation.

Operate the robot only when it is safe to do so.



Spark 2

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